Analytics Take-Home Exercise

Build a dashboard with an AI feature, and recommend what to do | Consumer Intelligence & Analytics

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Thanks for taking the time. We run this same exercise with every candidate so we can compare fairly, so please treat the light ambiguity below as intentional. We expect about **half a day** of focused work, so please don't spend more than that. Polish is not what we're looking for.

The dataset

Attached is `TYLY_data_sample.xlsx`, a dummy extract modeled on our real customer, sales, and inventory data (all values are fabricated). It has three sheets:

- **Column Definitions** explains every column. Please read it closely before you dive in.
- **Data** holds sales at the transaction-line grain, for this year (FY26) and last year (FY25).
- **Inventory** holds weekly stock by product family, for this year and last year.

How you interpret the structure is part of the exercise.

Business context to keep in mind

These are the goals we care about, offered as context rather than a checklist:

- We monitor both **year-over-year** and **trailing-week** performance.
- Our top priority is **acquiring new Gen Z customers** and **retaining customers** over time.
- We like to **surface winning products** worth sharing across the business.
- We want stock in the right place, so **inventory** is included.

The ask

Use any generative-AI tool you like (for example Claude, ChatGPT, Copilot, or Cursor). Vibecode it. There is no single right answer, and we want to see your genuine approach rather than a template. Please build two things:

1. A dashboard or small web app. It should show how the business is performing and include at least one **AI-powered feature** that someone on the business side would use every week. Some examples are an auto-written weekly summary, a chatbot that answers questions about the data, or an alert when something unusual happens. Your own idea is welcome too. A rough working prototype is enough. You choose the tools, the format, and the metrics. **You don't need a paid AI API.** A free tier, a local model, AI-written content you generate once and show in the dashboard, or a mock that shows where the AI plugs in are all fine. We look at the design and how you keep its answers accurate, not whether it calls a live model.

2. A short deck (5 to 8 slides) that covers:

- your key findings;
- your top **2 to 3 recommendations**, ranked by priority, each with the reason, the impact you expect, and any data you would want that is not in this file;
- how you built it: the AI tools and key prompts you used, how you checked the AI's output, and how your AI feature keeps its answers accurate.

What to send back

- A **link** to your dashboard or app. If you can't share a working link, send the files or a short screen recording instead.
- Your **deck**.

How we will look at it

We are looking for your thought process, creativity, and instinct for analytics and innovation. Correct, decision-ready insight matters more than the number of charts. If something in the data surprises you, tell us. In the follow-up conversation we will ask you to present your deck and walk us through your dashboard.

Please submit your work by the deadline in your invitation email. Questions are welcome at any point. We look forward to seeing what you build.